Chapter 6. Building and Applying Logistic Regression Models

Model selection, diagnostics, conditional inference

Contents

6.1 Strategies in model selection **2**
- 6.1.1 Horseshoe crab example revisited . . . 2
- 6.1.2 Stepwise procedures . . . 2
- 6.1.3 Backward elimination for the horseshoe crab example . . . 2
- 6.1.4 AIC, model selection, and the "correct" model . . . 3
- 6.1.5 Using causal hypotheses to guide model building . . . 4
- 6.1.6 New model-building strategies for data mining . . . 4

6.2 Logistic regression diagnostics **4**
- 6.2.1 Pearson, deviance, and standardized residuals . . . 4
- 6.2.2 Heart disease example . . . 5
- 6.2.3 Graduate admissions example . . . 6
- 6.2.4 Influence diagnostics for logistic regression . . . 6
- 6.2.5 Summarizing predictive power: R and R-squared measures . . . 7
- 6.2.6 Summarizing predictive power: classification tables and ROC curves . . . 7

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables **8**
- 6.3.1 Using logit models to test conditional independence . . . 8
- 6.3.2 Cochran–Mantel–Haenszel test of conditional independence . . . 9
- 6.3.3 Multicenter clinical trial example . . . 10
- 6.3.4 Estimation of common odds ratio . . . 10
- 6.3.5 Testing homogeneity of odds ratios . . . 11
- 6.3.6 Summarizing heterogeneity in odds ratios . . . 12

6.1 Strategies in model selection

Motivation. We want a model complex enough to fit but simple enough to interpret.

Confirmatory studies compare a few prespecified models; exploratory searches only generate hypotheses. Look first at each predictor's marginal effect. Guidelines: at least 10 outcomes of each type per predictor; beware multicollinearity.

6.1.1 Horseshoe crab example revisited

Example 6.1 (Horseshoe crabs: all main effects)

Crabs: $y = 1$ if a satellite; predictors color C (4 categories), spine S (3), weight, width W. The main-effects model (dummies c_1, c_2, c_3, s_1, s_2) has LR 40.6 against the null ($df = 7$, $P < 0.0001$), yet no single effect is clearly significant:

Parameter	Estimate	SE	Parameter	Estimate	SE
Intercept	−9.273	3.838	color 2	1.506	0.567
weight	0.826	0.704	color 3	1.120	0.593
width	0.263	0.195	spine 1	−0.400	0.503
color 1	1.609	0.936	spine 2	−0.496	0.629

Multicollinearity (corr(weight, width) = 0.887): keep W, C, S. Models are named by highest-order terms ($C + S + W$, $C + S{*}W$, ...), main effects included.

6.1.2 Stepwise procedures

Forward selection adds the term with the smallest P-value (*stepwise* also re-tests earlier terms); *backward elimination* removes the term with the largest P-value until every deletion significantly worsens the fit. Add or drop factors as a whole. Backward is preferred (forward may stop early through one low-power test). Selected effects are overestimated; a variable central to the study may be kept anyway.

6.1.3 Backward elimination for the horseshoe crab example

With a continuous predictor G^2 is not χ^2, but deviance differences of nested models are approximately χ^2.

Example 6.2 (Horseshoe crabs: backward elimination)

Selected rows of the book's Table 6.2:

	Model	G^2	df	AIC	Comparison	Difference
1	$C{*}S{*}W$	170.44	152	212.4	–	–
2	$C{*}S + C{*}W + S{*}W$	173.68	155	209.7	(2)–(1)	3.2 ($df = 3$)
3c	$C{*}S + C{*}W$	173.69	157	205.7	(3c)–(2)	0.0 ($df = 2$)
4b	$W + C{*}S$	177.61	160	203.6	(4b)–(3c)	3.9 ($df = 3$)
5	$C + S + W$	186.61	166	200.6	(5)–(4b)	9.0 ($df = 6$)
6c	$C + W$	187.46	168	197.5	(6c)–(5)	0.8 ($df = 2$)
7a	C	212.06	169	220.1	(7a)–(6c)	24.5 ($df = 1$)
7b	W	194.45	171	198.5	(7b)–(6c)	7.0 ($df = 3$)
8	$C = \text{dark} + W$	187.96	170	194.0	(8)–(6c)	0.5 ($df = 2$)
9	None	225.76	172	227.8	(9)–(8)	37.8 ($df = 2$)

All interactions drop out, then S; removing C from $C + W$ costs 7.0 ($df = 3$, $P = 0.07$) and removing W costs 24.5. A single dark/not-dark dummy loses almost nothing: final model width + dark.

6.1.4 AIC, model selection, and the "correct" model

No model is exactly true; a parsimonious one may estimate better. AIC measures the expected closeness of the fit to the truth.

Definition (AIC)

For a model with p parameters and maximized log likelihood L_M, AIC $= -2(L_M - p) = -2L_M + 2p$; smaller is better.

Proposition 6.3 (AIC and the adjusted deviance)

Fit models to the same data with N grouped observations (saturated model N parameters, $df = N - p$). Then

$$\text{AIC} = [G^2 - 2\,df] + \text{const},$$

where $-2L_S + 2N$ is common to all models; so AIC orders models as $G^2 - 2\,df$ does.

Proof. $G^2 = -2(L_M - L_S)$ gives $-2L_M = G^2 - 2L_S$; with $p = N - df$, AIC $= [G^2 - 2\,df] + (2N - 2L_S)$, and the last term depends only on the data. ■

Theorem 6.4 (Akaike's bias correction)

Let $Y_1, \ldots, Y_n$ be i.i.d. with density $f(\cdot; \theta_0)$ belonging to a regular parametric model with p-dimensional parameter, $\hat\theta$ the MLE and $\ell(\theta) = \sum_i \log f(Y_i; \theta)$. Let $Y_1^*, \ldots, Y_n^*$ be an independent copy of the data and $\ell^*(\theta)$ its log likelihood. Then, to first order,

$$E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p.$$

So AIC $= -2\ell(\hat\theta) + 2p$ estimates the expected out-of-sample deviance $-2E[\ell^*(\hat\theta)]$ (i.e. $2n$ times the Kullback–Leibler discrepancy, up to a constant).

Proof. [Supplement] Let I be the per-observation Fisher information, so $\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, I^{-1})$.

Step 1. Expanding about $\hat\theta$ ($\ell'(\hat\theta) = 0$), $\ell(\theta_0) \approx \ell(\hat\theta) - \frac{n}{2}(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0)$; since $n(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0) \xrightarrow{d} \chi^2_p$, $E[\ell(\hat\theta)] \approx E[\ell(\theta_0)] + p/2$.

Step 2. Expanding ℓ^* about θ_0, and using that $\ell^{*\prime}(\theta_0)$ has mean zero and is independent of $\hat\theta$,

$$E[\ell^*(\hat\theta)] \approx E[\ell^*(\theta_0)] - \tfrac{n}{2} E[(\hat\theta-\theta_0)^\top I(\hat\theta-\theta_0)] = E[\ell^*(\theta_0)] - \tfrac{p}{2}.$$

Step 3. $\ell(\theta_0)$ and $\ell^*(\theta_0)$ are equal in distribution, so subtracting gives $E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p$, i.e. $E[-2\ell(\hat\theta)+2p] \approx -2E[\ell^*(\hat\theta)]$. Finally $-2E[\ell^*(\theta)]/n = 2\,\mathrm{KL}(f_{\theta_0}\|f_\theta) - 2E\log f(Y;\theta_0)$, whose last term is model-free. ■

Example 6.5 (Horseshoe crabs: AIC)

AIC is smallest for $C+W$ (197.5) among C, S, W models, smaller still for the dark-indicator model (194.0), itself suggested by the $C+W$ fit.

6.1.5 Using causal hypotheses to guide model building

A time ordering suggests a sequence of logit models, each response explained by earlier variables.

Example 6.6 (Divorce and pre/extramarital sex)

G = gender, P = premarital sex, E = extramarital sex, M = marital status:

	Women				Men			
PMS	Yes		No		Yes		No	
EMS	Yes	No	Yes	No	Yes	No	Yes	No
Divorced	17	54	36	214	28	60	17	68
Still married	4	25	4	322	11	42	4	130

For $G \to P \to E \to M$: P on none, $G^2 = 75.3$ ($df = 1$; G–P odds ratio 0.27); E on none / P / $G+P$, $G^2 = 48.9/2.9/0.0$ ($df = 3/2/1$), weak evidence of a direct G effect; M on $E+P$ / $E{*}P$ / $E{*}P+G$, $G^2 = 18.2/5.2/0.7$ ($df = 5/4/3$), so $E \times P$ is needed. The causal reasoning led close to a well-fitting model.

6.1.6 New model-building strategies for data mining

In huge data sets nearly every effect is significant; predictive-power summaries (Section 6.2) matter more.

6.2 Logistic regression diagnostics

Motivation. Residuals and influence measures show where a model fails.

6.2.1 Pearson, deviance, and standardized residuals

Let $y_i \sim \mathrm{bin}(n_i, \pi_i)$ at setting $i = 1, \ldots, N$, with fitted $\hat\pi_i$ and leverage $\hat h_i$, the ith diagonal element of $\hat W^{1/2} X (X^\top \hat W X)^{-1} X^\top \hat W^{1/2}$, $\hat W = \mathrm{diag}(n_i \hat\pi_i(1-\hat\pi_i))$.

Definition (Residuals for binomial GLMs)

(a) Pearson residual:

$$e_i = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}. \tag{6.1}$$

(b) Standardized (adjusted) Pearson residual: $r_i = e_i/\sqrt{1-\hat{h}_i}$.
(c) Deviance residual: $\sqrt{d_i}\ \mathrm{sgn}(y_i - n_i\hat{\pi}_i)$, with $d_i = 2[y_i \log \frac{y_i}{n_i\hat{\pi}_i} + (n_i - y_i)\log \frac{n_i - y_i}{n_i - n_i\hat{\pi}_i}]$.

Thus $X^2 = \sum_i e_i^2$ and $G^2 = \sum_i d_i$.

Theorem 6.7 (Approximate distribution of residuals)

Suppose the logistic model holds with p parameters and the n_i are large. Then, approximately,
(a) $\mathrm{Var}(y_i - n_i\hat{\pi}_i) \approx n_i\pi_i(1-\pi_i)(1-h_i)$, so $\mathrm{Var}(e_i) \approx 1 - h_i < 1$ while r_i is approximately $N(0,1)$;
(b) $E(\sum_i e_i^2) \approx N - p = df$, so $E(\sum_i e_i^2)/N \approx df/N < 1$.

Proof. [Supplement] Write $\mu = (n_i\pi_i)$, $W = \mathrm{diag}(n_i\pi_i(1-\pi_i)) = \mathrm{Cov}(y)$, $A = X^\top W X$.

Step 1. The logit likelihood equations are $X^\top(y - \hat{\mu}) = 0$, and $\partial\mu/\partial\beta^\top = WX$, so $\hat{\mu} - \mu \approx WX(\hat{\beta} - \beta)$ and $0 \approx X^\top(y-\mu) - A(\hat{\beta} - \beta)$, i.e. $\hat{\beta} - \beta \approx A^{-1}X^\top(y - \mu)$.

Step 2. Hence $y - \hat{\mu} \approx (I - WXA^{-1}X^\top)(y - \mu)$ and

$$\mathrm{Cov}(y - \hat{\mu}) \approx (I - WXA^{-1}X^\top)\, W\, (I - XA^{-1}X^\top W) = W - WXA^{-1}X^\top W.$$

Step 3. So $\mathrm{Cov}(W^{-1/2}(y - \hat{\mu})) \approx I - H$, $H = W^{1/2}XA^{-1}X^\top W^{1/2}$ with diagonal (h_i). The components of $W^{-1/2}(y - \hat{\mu})$ are the e_i (with π_i for $\hat{\pi}_i$, a negligible change), proving (a); normality follows from that of y, and $r_i = e_i/\sqrt{1-h_i}$ has unit variance.

Step 4. H is a projection of rank p, so $E\sum e_i^2 \approx \mathrm{tr}(I - H) = N - p$. ∎

With $n_i = 1$ (continuous predictors) residuals take two values and are uninformative; group observations where possible.

6.2.2 Heart disease example

Example 6.8 (Blood pressure and heart disease)

Framingham men, eight blood-pressure categories (scores $111.5, \ldots, 191.5$), response: heart disease within six years.

Blood pressure	Sample size	Heart disease	Fitted		Std. residual	
			Indep.	Linear	Indep.	Linear
< 117	156	3	10.8	5.2	−2.62	−1.11
117–126	252	17	17.4	10.6	−0.12	2.37
127–136	284	12	19.7	15.1	−2.02	−0.95
137–146	271	16	18.8	18.1	−0.74	−0.57
147–156	139	12	9.6	11.6	0.84	0.13
157–166	85	8	5.9	8.9	0.93	−0.33
167–186	99	16	6.9	14.2	3.76	0.65
> 186	43	8	3.0	8.4	3.07	−0.18

Independence fits poorly ($G^2 = 30.0$, $df = 7$), residuals increasing in x; the linear logit model removes the trend ($G^2 = 5.9$, $df = 6$), leaving only $r_2 = 2.37$.

6.2.3 Graduate admissions example

Example 6.9 (Admissions at the University of Florida)

23 departments: no gender effect, $\operatorname{logit}\pi_{ik} = \alpha + \beta_k^D$, fits rather poorly ($G^2 = 44.7$, $df = 23$). Large standardized residuals for (female, admitted): astronomy 2.87 (6 of 6 females, 3 of 11 males), geography 2.17, psychology -2.27; without these three, $G^2 = 24.4$, $df = 20$. A gender effect barely helps ($G^2 = 42.4$, $df = 22$; deviations go both ways). Its conditional odds ratio 1.19 (favouring females) versus marginal 0.94 is Simpson's paradox.

Proposition 6.10 (Residuals under conditional independence)

In a $2 \times 2 \times K$ table with $y_{ik} \sim \text{bin}(n_{ik}, \pi_{ik})$, $\operatorname{logit}\pi_{ik} = \alpha + \beta_k$ has $\hat\pi_{1k} = \hat\pi_{2k} = y_{+k}/n_{+k}$ and $y_{1k} - n_{1k}\hat\pi_{1k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$. So the two standardized residuals in stratum k are equal and opposite; the Pearson residuals need not be.

Proof.

Step 1. The likelihood equation for β_k is $\sum_i y_{ik} = \sum_i n_{ik}\hat\pi_{ik}$, and $\hat\pi_{1k} = \hat\pi_{2k}$, so $\hat\pi_{ik} = y_{+k}/n_{+k}$.

Step 2. Then $y_{1k} - n_{1k}\hat\pi_{1k} = (n_{2k}y_{1k} - n_{1k}y_{2k})/n_{+k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$.

Step 3. Negatives of each other have the same standard error, so the standardized residuals are $\pm$ the same number; the Pearson residuals divide instead by $\sqrt{n_{ik}\hat\pi_{ik}(1-\hat\pi_{ik})}$, which differs for $i = 1, 2$ when $n_{1k} \neq n_{2k}$. ■

6.2.4 Influence diagnostics for logistic regression

Deleting an observation equals adding a parameter that fits it perfectly. Leverage depends on $\hat W$, so an extreme x_i with $\hat\pi_i$ near 0 or 1 can have small leverage. Influence measures: *Dfbeta* (change in $\hat\beta_j$ on deleting i, over its SE); *confidence interval displacement* c_i; the drops ΔX_i^2, ΔG_i^2.

Theorem 6.11 (One-step influence approximations)

Let $A = X^\top \hat W X$, $w_i = n_i\hat\pi_i(1-\hat\pi_i)$, $h_i = w_i x_i^\top A^{-1} x_i$, and let $\hat\beta_{(i)}$ be the one-step Newton approximation (started from $\hat\beta$) to the estimate without observation i. Then

$$\hat\beta - \hat\beta_{(i)} = \frac{A^{-1}x_i\,(y_i - n_i\hat\pi_i)}{1-h_i}, \qquad c_i = (\hat\beta - \hat\beta_{(i)})^\top A(\hat\beta - \hat\beta_{(i)}) = \frac{e_i^2 h_i}{(1-h_i)^2},$$

and the score statistic for the extra parameter that fits observation i perfectly equals $r_i^2 = e_i^2/(1-h_i)$, the usual approximation to ΔX_i^2.

Proof. [Supplement]

Step 1. The full score vanishes at $\hat\beta$, so the reduced-data score is $U_{(i)}(\hat\beta) = -x_i(y_i - n_i\hat\pi_i)$, with information $A - w_i x_i x_i^\top$; one Newton step gives $\hat\beta_{(i)} = \hat\beta + (A - w_i x_i x_i^\top)^{-1} U_{(i)}(\hat\beta)$.

Step 2. By Sherman–Morrison,

$$(A - w_i x_i x_i^\top)^{-1} x_i = A^{-1}x_i + \frac{w_i A^{-1}x_i x_i^\top A^{-1}x_i}{1 - w_i x_i^\top A^{-1} x_i} = \frac{A^{-1}x_i}{1-h_i},$$

giving $\hat\beta - \hat\beta_{(i)}$.

Step 3. Then $c_i = (y_i - n_i\hat\pi_i)^2 x_i^\top A^{-1}x_i/(1-h_i)^2 = (y_i - n_i\hat\pi_i)^2 h_i/\{w_i(1-h_i)^2\} = e_i^2 h_i/(1-h_i)^2$.

Step 4. Add γ with covariate $\mathbf{e}_i$. At $(\hat\beta, 0)$, $U_\gamma = y_i - n_i\hat\pi_i$, $I_{\gamma\gamma} = w_i$, $I_{\gamma\beta} = w_i x_i^\top$, $I_{\beta\beta} = A$; the efficient information is $w_i - w_i^2 x_i^\top A^{-1} x_i = w_i(1-h_i)$, so the score statistic is $(y_i - n_i\hat\pi_i)^2/\{w_i(1-h_i)\} = r_i^2$. Fitting γ equals deleting i, so this approximates the drop in X^2 (and G^2). ■

Example 6.12 (Heart disease: influence)

Linear logit model of Example 6.8: category 2 has by far the largest Dfbeta (-1.14), c (2.26), ΔX^2 (5.64), ΔG^2 (5.04); all others are below 1.4.

6.2.5 Summarizing predictive power: R and R-squared measures

No R^2 analog is universally useful. Proposals:

(a) the correlation $r(y, \hat\pi)$ (crabs: 0.402 for W, 0.452 for $C+W$, 0.526 for $C{*}S{*}W$, the last inflated by its many parameters);

(b) $1 - \sum(y_i - \hat\pi_i)^2 / \sum(y_i - \bar y)^2$;

(c) likelihood-based measures.

Proposition 6.13 (Likelihood R^2 measures)

Let $L_0 \le L_M \le L_S$ be the maximized log likelihoods of the null (intercept-only), the given and the saturated models. Then $(L_M - L_0)/(L_S - L_0) \in [0,1]$, and:
(a) for n ungrouped Bernoulli observations, $L_S = 0$ and the measure reduces to McFadden's $D = (L_0 - L_M)/L_0$, where $L_0 = n[\bar y \log \bar y + (1-\bar y)\log(1-\bar y)]$;
(b) for N grouped binomial counts it equals $D^* = \{G^2(0) - G^2(M)\}/G^2(0)$.

Proof. The models are nested, so $L_0 \le L_M \le L_S \le 0$, giving the range.

(a) The saturated fit $\hat\pi_i = y_i \in \{0,1\}$ makes every term $y_i \log\hat\pi_i + (1-y_i)\log(1-\hat\pi_i)$ zero, so $L_S = 0$; the null fit $\hat\pi_i = \bar y$ gives L_0.

(b) $G^2(M) = -2(L_M - L_S)$ and $G^2(0) = -2(L_0 - L_S)$; take the ratio. ■

D^* is misleading: it is 1 for any perfect fit and tends to 1 whenever the model holds; use D on ungrouped data.

6.2.6 Summarizing predictive power: classification tables and ROC curves

A *classification table* cross-classifies y with $\hat y = \mathbb{1}\{\hat\pi_i > \pi_0\}$ (sensitivity, specificity); it discards information and depends on the arbitrary π_0.

Definition (ROC curve and concordance index)

The *ROC curve* plots sensitivity against 1−specificity as π_0 ranges over $[0,1]$. With n_1, n_0 the numbers of 1's and 0's, the *concordance index*

$$c = \frac{1}{n_1 n_0} \sum_{i:\, y_i=1} \sum_{j:\, y_j=0} \Big[\mathbb{1}\{\hat\pi_i > \hat\pi_j\} + \tfrac{1}{2}\mathbb{1}\{\hat\pi_i = \hat\pi_j\} \Big]$$

is the proportion of (case, non-case) pairs in which the case has the larger fitted probability.

Proposition 6.14 (Area under the ROC curve)

The area under the empirical ROC curve (with linear interpolation across ties) equals the concordance index c.

Proof. [Supplement]

Step 1. Lower π_0 from 1 to 0. Passing a distinct fitted value v moves the curve from (a, b) by $(m_0(v)/n_0, m_1(v)/n_1)$, where $m_y(v)$ counts observations with response y and $\hat{\pi} = v$, and $b = \#\{i : y_i = 1, \hat{\pi}_i > v\}/n_1$.

Step 2. The trapezoid added has area

$$\frac{m_0(v)}{n_0}\Big(b + \frac{m_1(v)}{2n_1}\Big) = \frac{1}{n_0 n_1} \sum_{j:\, y_j=0,\ \hat{\pi}_j=v} \Big[\#\{i : y_i = 1, \hat{\pi}_i > v\} + \tfrac{1}{2}\#\{i : y_i = 1, \hat{\pi}_i = v\}\Big].$$

Step 3. Summing over v runs over every non-case j once, giving c. ■

$c = 0.5$ is random guessing. Crabs: $c = 0.639$ (color), 0.742 (width), 0.771 (width and color).

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables

Motivation. Logit models, the CMH test and the Mantel–Haenszel estimator compare two treatments across strata.

Example 6.15 (Eight-center clinical trial)

Drug vs. control cream for curing an infection (μ_{11k}, $\mathrm{Var}(n_{11k})$: null hypergeometric moments of drug successes):

	Drug		Control				
Center	Succ.	Fail.	Succ.	Fail.	Odds ratio	μ_{11k}	$\mathrm{Var}(n_{11k})$
1	11	25	10	27	1.19	10.36	3.79
2	16	4	22	10	1.82	14.62	2.47
3	14	5	7	12	4.80	10.50	2.41
4	2	14	1	16	2.29	1.45	0.70
5	6	11	0	12	∞	3.52	1.20
6	1	10	0	10	∞	0.52	0.25
7	1	4	1	8	2.0	0.71	0.42
8	4	2	6	1	0.33	4.62	0.62

6.3.1 Using logit models to test conditional independence

For binary Y, binary X ($x_1 = 1$, $x_2 = 0$) and control Z, let $\pi_{ik} = P(Y = 1 \mid X = i, Z = k)$ and

$$\operatorname{logit} \pi_{ik} = \alpha + \beta x_i + \beta_k^Z, \qquad i = 1, 2,\ k = 1, \ldots, K, \tag{6.2}$$

with common conditional odds ratio e^{β}. Conditional independence $H_0 : \beta = 0$ is tested by Wald or by LR against

$$\operatorname{logit} \pi_{ik} = \alpha + \beta_k^Z, \tag{6.3}$$

both $df = 1$. The goodness-of-fit test of (6.3) ($df = K$) is less powerful when the association is similar across strata.

6.3.2 Cochran–Mantel–Haenszel test of conditional independence

Given both margins of each partial table, n_{11k} determines the table.

Proposition 6.16 (Null hypergeometric moments)

Under conditional independence, given the margins of partial table k, n_{11k} is hypergeometric with

$$\mu_{11k} = E(n_{11k}) = \frac{n_{1+k}n_{+1k}}{n_{++k}}, \qquad \mathrm{Var}(n_{11k}) = \frac{n_{1+k}n_{2+k}n_{+1k}n_{+2k}}{n_{++k}^2(n_{++k}-1)}.$$

Proof. [Supplement] Drop k, let $n = n_{++}$, $q = n_{+1}/n$. Under H_0 the n_{+1} successes are a random subset of the n subjects. With I_s the success indicator of row-1 subject s, $n_{11} = \sum_s I_s$, $P(I_s = 1) = q$ and $P(I_s = I_t = 1) = n_{+1}(n_{+1}-1)/\{n(n-1)\}$ for $s \neq t$.

Step 1. $E(n_{11}) = n_{1+}q = n_{1+}n_{+1}/n$.

Step 2. $\mathrm{Var}(I_s) = q(1-q)$ and $\mathrm{Cov}(I_s, I_t) = -q(1-q)/(n-1)$, so

$$\mathrm{Var}(n_{11}) = n_{1+}q(1-q)\Big[1 - \frac{n_{1+}-1}{n-1}\Big] = n_{1+}q(1-q)\frac{n-n_{1+}}{n-1} = \frac{n_{1+}n_{2+}n_{+1}n_{+2}}{n^2(n-1)}.$$

■

Definition (CMH statistic)

$$\mathrm{CMH} = \frac{\left[\sum_k (n_{11k} - \mu_{11k})\right]^2}{\sum_k \mathrm{Var}(n_{11k})}, \tag{6.4}$$

approximately χ^2_1 under H_0. Cochran's version uses the variance $n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n_{++k}^3$.

Deviations accumulate when the association has the same direction in every stratum. The hypergeometric form also covers retrospective studies and randomized trials.

Theorem 6.17 (CMH as a score test for the logit model)

For model (6.2) with independent binomial rows:
(a) the (unconditional) score statistic for $H_0 : \beta = 0$ is Cochran's statistic;
(b) the score statistic for β based on the conditional likelihood given all row and column margins is exactly CMH. So CMH and Cochran differ only by the factors $(n_{++k}-1)/n_{++k}$ in the variance, and both are asymptotically equivalent to the Wald and LR tests of $\beta = 0$.

Proof. [Supplement] Absorb α: $\mathrm{logit}\,\pi_{ik} = \beta x_i + \gamma_k$.

Step 1. (Score.) From $L = \sum_{ik}[y_{ik}(\beta x_i + \gamma_k) - n_{ik}\log(1 + e^{\beta x_i + \gamma_k})]$, $\partial L/\partial\beta = \sum_k (y_{1k} - n_{1k}\pi_{1k})$ and $\partial L/\partial\gamma_k = \sum_i (y_{ik} - n_{ik}\pi_{ik})$. Under H_0, $\hat\pi_{1k} = \hat\pi_{2k} = n_{+1k}/n_{++k}$ ($y_{ik} = n_{i1k}$), so $U = \sum_k (n_{11k} - \mu_{11k})$.

Step 2. (Information.) With $v_k = \pi_k(1-\pi_k)$: $I_{\beta\beta} = \sum_k n_{1+k}v_k$, $I_{\beta\gamma_k} = n_{1+k}v_k$, $I_{\gamma_k\gamma_k} = n_{++k}v_k$, $I_{\gamma_k\gamma_l} = 0$ ($k \neq l$). The efficient information is

$$I_{\beta\beta} - \sum_k \frac{I^2_{\beta\gamma_k}}{I_{\gamma_k\gamma_k}} = \sum_k v_k \frac{n_{1+k}n_{2+k}}{n_{++k}};$$

with $\hat v_k = n_{+1k}n_{+2k}/n_{++k}^2$, U^2 over this is Cochran's statistic: (a).

Step 3. (Conditional score.) Given all margins,

$$P(n_{11k} = t_k,\ k = 1, \ldots, K) = \prod_k \frac{c_k(t_k)e^{\beta t_k}}{\sum_u c_k(u)e^{\beta u}},$$

a one-parameter exponential family with statistic $T = \sum_k n_{11k}$, score $T - E_\beta(T)$ and information $\mathrm{Var}_\beta(T)$. At $\beta = 0$ these are $\sum_k (n_{11k} - \mu_{11k})$ and $\sum_k \mathrm{Var}(n_{11k})$ (Proposition 6.16), so the statistic is CMH: (b). The last claim is standard score/Wald/LR asymptotics. ■

6.3.3 Multicenter clinical trial example

Example 6.18 (Clinical trial: tests)

All centers but the last show a positive association: CMH = 6.38 ($P = 0.012$); model (6.2) gives $\hat{\beta} = 0.777$ (SE 0.307), Wald 6.42, LR 6.67.

6.3.4 Estimation of common odds ratio

Under (6.2) the ML estimate of $\theta = e^\beta$ is $e^{\hat{\beta}}$; non-model alternatives are Woolf's estimator and the Mantel–Haenszel estimator.

Definition (Mantel–Haenszel estimator)

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k} n_{22k}/n_{++k}}{\sum_k n_{12k} n_{21k}/n_{++k}} = \frac{\sum_k R_k}{\sum_k S_k} = \frac{R}{S}. \tag{6.5}$$

Theorem 6.19 (Unbiased estimating equation for $\hat{\theta}_{MH}$)

If the rows of each partial table are independent binomials and the odds ratio equals θ in every stratum, then $E(R - \theta S) = 0$. Consequently $\hat{\theta}_{MH}$ is consistent both as $n \to \infty$ with K fixed and as $K \to \infty$ with bounded strata sizes.

Proof. [Supplement]

Step 1. With independent $n_{11k} \sim \mathrm{bin}(n_{1+k}, \pi_{1k})$, $n_{21k} \sim \mathrm{bin}(n_{2+k}, \pi_{2k})$: $E(n_{11k} n_{22k}) = n_{1+k} n_{2+k} \pi_{1k}(1 - \pi_{2k})$ and $E(n_{12k} n_{21k}) = n_{1+k} n_{2+k}(1 - \pi_{1k})\pi_{2k}$.

Step 2. Since $\theta = \pi_{1k}(1 - \pi_{2k})/\{(1 - \pi_{1k})\pi_{2k}\}$, $E(R_k - \theta S_k) = 0$; sum over k.

Step 3. Under either scheme, with $m = n$ or $m = K$, R/m and S/m are averages of independent terms with bounded variances, so $(R - \theta S)/m \xrightarrow{p} 0$ while S/m stays bounded away from 0 (nondegenerate strata). Hence $\hat{\theta}_{MH} - \theta = (R - \theta S)/S \xrightarrow{p} 0$. ■

Theorem 6.20 (Unconditional ML breaks down for matched pairs)

In model (6.2) with one matched pair per stratum, let b (c) be the number of pairs in which only the first (only the second) member has $y = 1$. Then the unconditional ML estimator is $\hat{\beta} = 2\log(b/c)$, whereas $\hat{\theta}_{MH} = b/c$. As $K \to \infty$, $\log(b/c) \xrightarrow{p} \beta$, so $\hat{\beta} \xrightarrow{p} 2\beta$.

Proof. [Supplement]

Step 1. A concordant pair's likelihood is maximized at $\gamma_k = \pm\infty$ with value 1: no information on β.

Step 2. With $u = e^{\gamma_k}$, $\psi = e^\beta$, the two discordant types have likelihoods $\frac{u\psi}{(1+u\psi)(1+u)}$ and $\frac{u}{(1+u\psi)(1+u)}$, both maximized at $u = \psi^{-1/2}$ (log-derivative $1/u - \psi/(1 + u\psi) - 1/(1 + u) = 0 \iff u^2\psi = 1$), with values $\psi/(1 + \psi^{1/2})^2$ and $1/(1 + \psi^{1/2})^2$.

Step 3. The profile log likelihood $b\beta - 2(b+c)\log(1+e^{\beta/2})$ has derivative $b-(b+c)e^{\beta/2}/(1+e^{\beta/2})$, zero at $e^{\beta/2} = b/c$: $\hat{\beta} = 2\log(b/c)$.

Step 4. $R_k = \frac{1}{2}$ in a b-type pair, $S_k = \frac{1}{2}$ in a c-type pair, both 0 otherwise, so $\hat{\theta}_{MH} = b/c$.

Step 5. Given discordance, a pair is b-type with probability $\pi_1(1-\pi_2)/\{\pi_1(1-\pi_2)+(1-\pi_1)\pi_2\} = \psi/(1+\psi)$, so $b/c \xrightarrow{p} \psi$. ■

The Robins–Breslow–Greenland variance estimate, valid under both asymptotic schemes, is

$$\hat{\sigma}^2(\log\hat{\theta}_{MH}) = \frac{\sum_k (n_{11k}+n_{22k})R_k/n_{++k}}{2R^2} + \frac{\sum_k (n_{12k}+n_{21k})S_k/n_{++k}}{2S^2} + \frac{\sum_k [(n_{11k}+n_{22k})S_k + (n_{12k}+n_{21k})R_k]/n_{++k}}{2RS}.$$

Example 6.21 (Clinical trial: common odds ratio)

For Example 6.15, $\hat{\theta}_{MH} = 2.13$, $\log\hat{\theta}_{MH} = 0.758$ ($\hat{\sigma} = 0.303$), 95% CI $(1.18, 3.87)$; the Wald and LR intervals for e^{β} are $(1.19, 3.97)$ and $(1.20, 4.02)$.

Neither $\hat{\theta}_{MH}$ nor CMH requires exactly equal odds ratios; both work well when the effects mostly share a direction.

6.3.5 Testing homogeneity of odds ratios

Homogeneity $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$ is model (6.2), tested by its G^2 or X^2 with $df = K-1$.

Proposition 6.22 (Fitted counts under a common odds ratio)

Fix the margins of stratum k and a value $\theta > 0$. There is a unique m_k with $\max(0, n_{1+k} + n_{+1k} - n_{++k}) < m_k < \min(n_{1+k}, n_{+1k})$ such that the table with $(1,1)$ entry m_k and the given margins has odds ratio θ:

$$\frac{m_k\,(n_{2+k} - n_{+1k} + m_k)}{(n_{1+k} - m_k)(n_{+1k} - m_k)} = \theta.$$

Proof. [Supplement] The other cells $n_{1+k} - m$, $n_{+1k} - m$, $n_{2+k} - n_{+1k} + m$ are all positive exactly on the stated open interval. There the left side is continuous, strictly increasing in m, tends to 0 at the lower end and to $+\infty$ at the upper end; by the intermediate value theorem there is exactly one root (of a quadratic in m). ■

Definition (Breslow–Day statistic)

With m_k from Proposition 6.22 at $\theta = \hat{\theta}_{MH}$ and $v_k \approx [1/m_k + 1/(n_{1+k} - m_k) + 1/(n_{+1k} - m_k) + 1/(n_{2+k} - n_{+1k} + m_k)]^{-1}$, $\mathrm{BD} = \sum_k (n_{11k} - m_k)^2/v_k$ is approximately χ^2_{K-1} under homogeneity.

Example 6.23 (Clinical trial: homogeneity)

$G^2 = 9.7$, $X^2 = 8.0$ ($df = 7$): no evidence against a common odds ratio.

6.3.6 Summarizing heterogeneity in odds ratios

Exactly equal odds ratios are unrealistic; a *random-effects* logit model takes the stratum log odds ratios from $N(\mu, \sigma^2)$ (Chapter 12).